

In Week 3 Lab, we'll explore Agentic RAG (Retrieval-Augmented Generation) systems with vector database (DB) to store a local knowledge base used by your AI agents. This week, we'll look at two potential approach to vector DBs:

- No-code workflow platform (make.com). Visual orchestration, no code at all. See below.
- Low-code approach using Chroma DB.

Week 3 Lab: Agentic AI + Vector Database ([make.com](https://www.make.com))

Concept (10-15 min, addresses the "show how it works" gap): Before building anything, walk through why a vector database is needed at all. Unstructured text (PDFs, articles, notes) can't be searched by keyword matching alone, semantic search finds meaning, not exact words. Demonstrate with one live example: ask Gemini the same question twice, once with no context, once with a relevant document chunk pasted in manually. Students see the answer quality difference before touching Make. This is the actual value the vector DB provides, retrieving the right chunk automatically so students don't do that pasting by hand.

Tools needed:

- Make.com free account
- Pinecone free account (vector database, has an official Make integration)
- Google Gemini AI connection (reuse API key from Week 1)

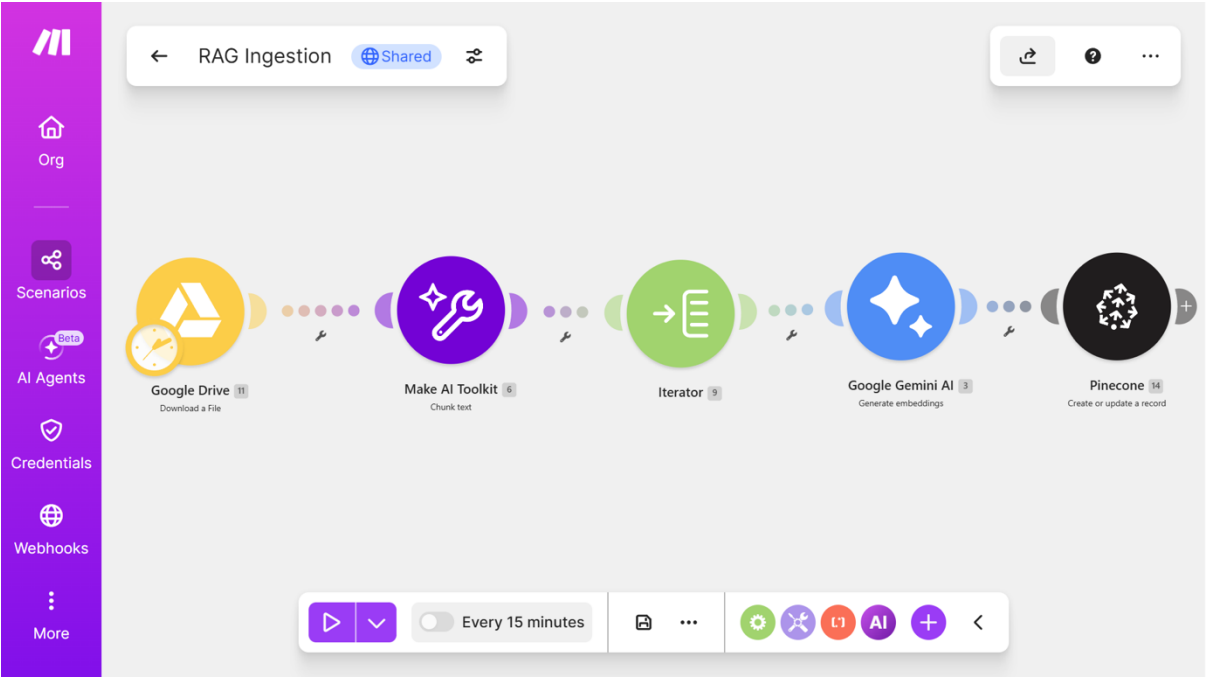
Data: .txt files (e.g. the provided [intelligent_agents_overview.txt](#)), https://drive.google.com/file/d/1OdufSCwTqM3AZDjfXhcx6qS6Hp8zcDI_/view?usp=sharing

Provided scenario (two parts):

- *Part 1: Ingestion pipeline*

https://drive.google.com/file/d/1TdTLq-nlfzHJ868n_khmZqKa1ZKZ7-Aj/view?usp=sharing

#	Module	Key settings
1	Google Drive → Download a File	gets raw text content
2	Make AI Toolkit → Chunk text	Text to chunk = Data from step 2; Chunk size = 200 ; Chunk overlap = 20
3	Iterator	Array = chunks output from step 3
4	Google Gemini AI → Generate embeddings	Output Dimensionality = 1024
5	Pinecone → Create or Update a Record	Namespace = week3-lab ; Record ID = {{Name}}-chunk-{{Bundle order position}} ; Values = embedding array from step 5; Metadata → text (String) = chunk text from step 4, source (String) = file Name

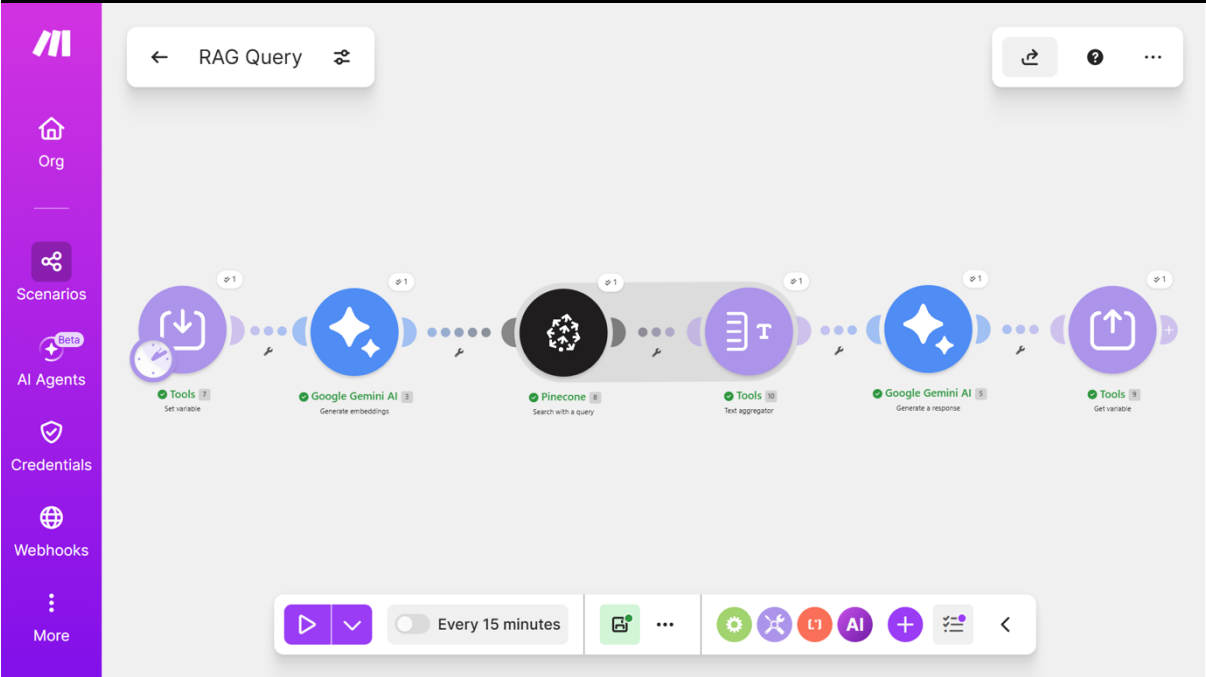


• Part 2: Query pipeline

<https://drive.google.com/file/d/1OzdMxch55ltgICarkvmujkMn-dY77V6E/view?usp=sharing>

#	Module	Key settings
1	Tools → Set variable	Variable name = question ; Value = typed test question
2	Google Gemini AI → Generate embeddings	Output Dimensionality = 1024 (must match Part 1 and the index)
3	Pinecone → Search with a query	Vector = embedding from step 2; Namespace = week3-lab ; Include Metadata = Yes ; Limit = 3 (this is the top-k value for the modification task)
4	Tools → Text aggregator	Source Module = step 3; Text = Metadata.text (mapped after one test run so the field expands); Row separator = \n\n---\n\n
5	Google Gemini AI → Generate a response	System: "answer only from context, say so if insufficient"; User: Context: {step 4} / Question: {step 1 question} / Answer using only the context above.

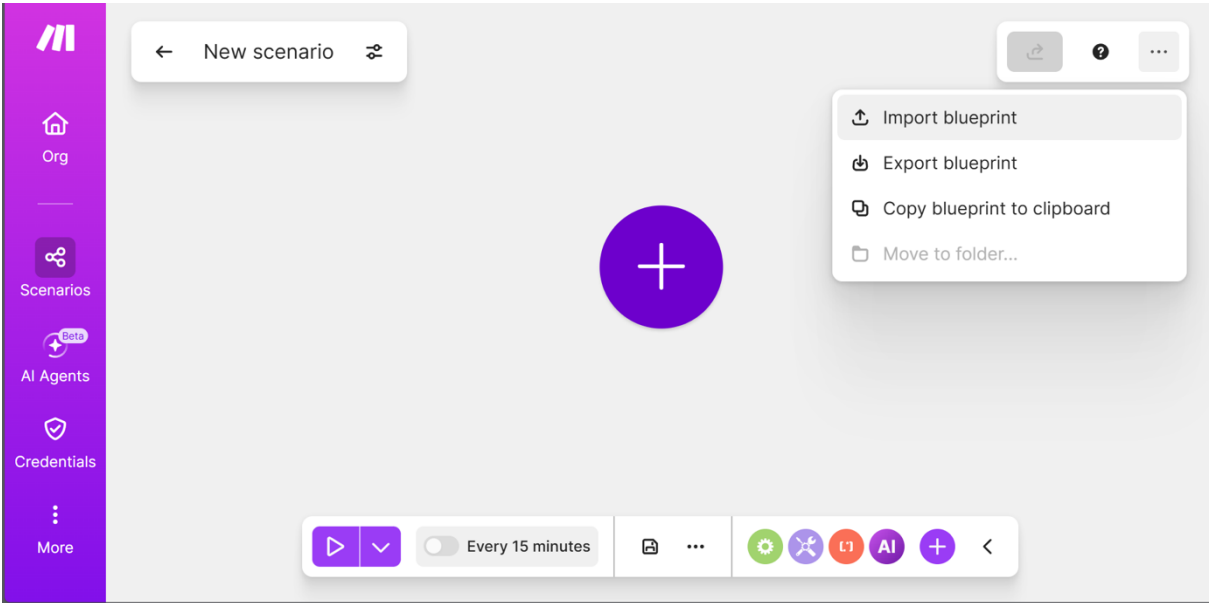
6	Tools → Get variable	retrieves the final answer for display when running inside Make
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Testing: Edit the text in **Set Variable** (step 1), click **Run once**, watch the answer appear at **step 6**.

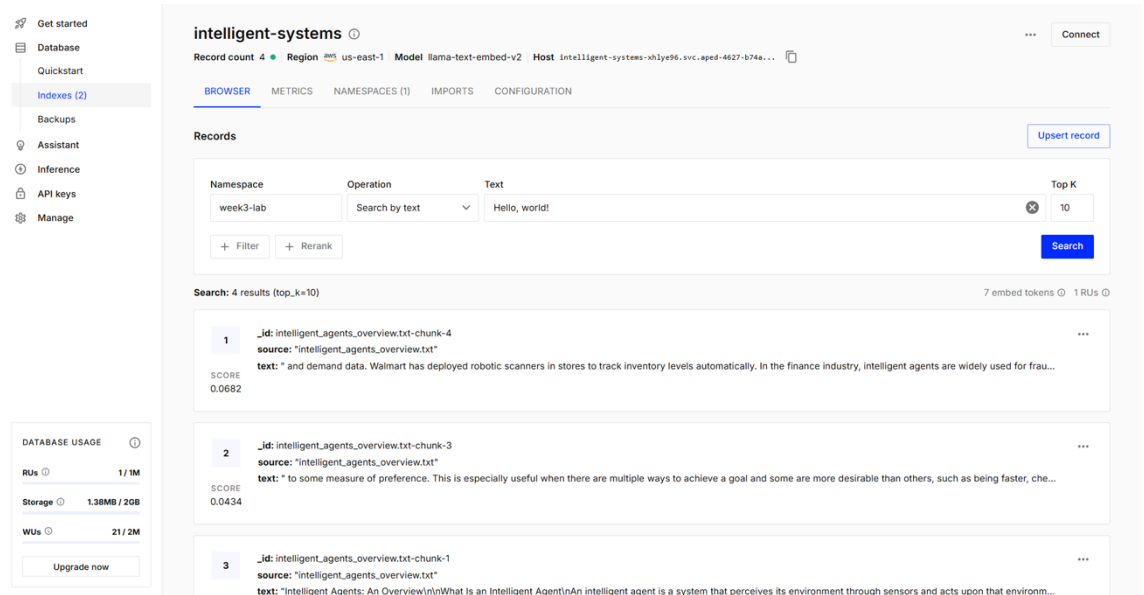
Steps for students:

- 1. Log in to Make, create two new scenarios and import the blueprint for each pipeline

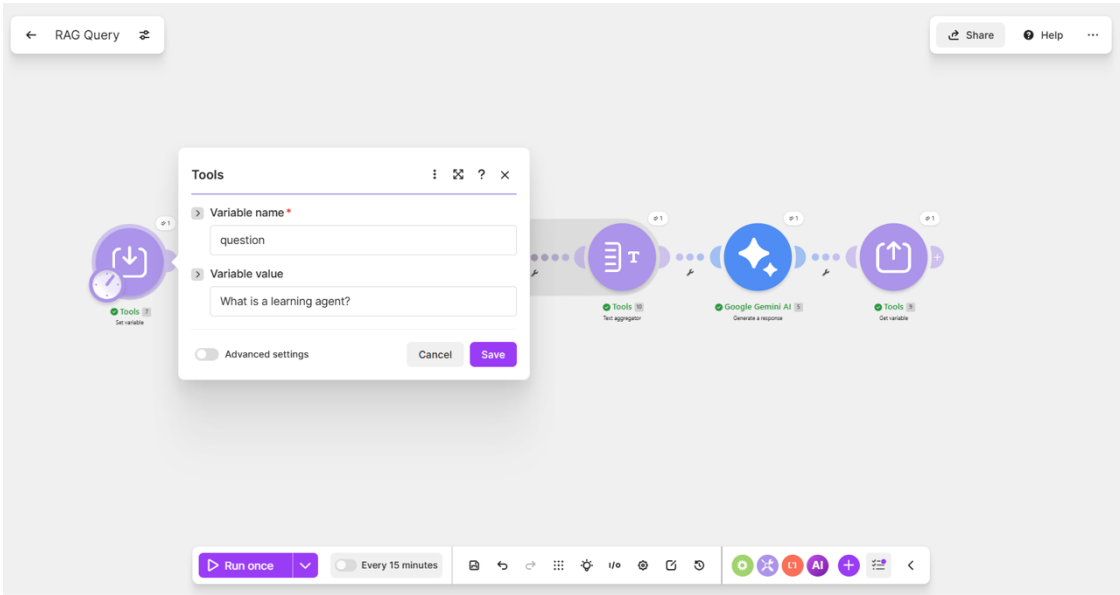


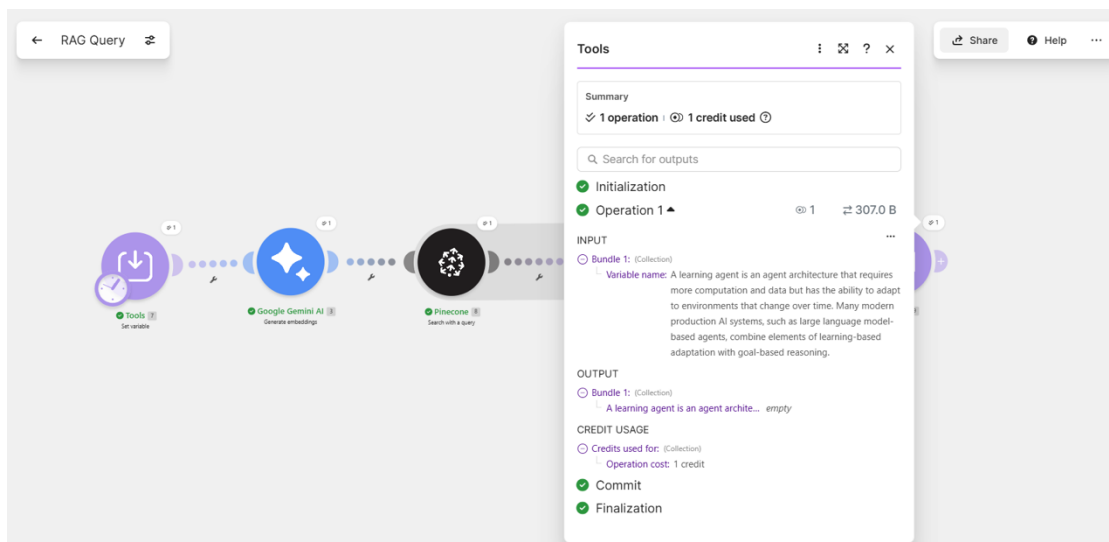
- 2. Connect their own Pinecone and Gemini accounts

3. Run Part 1 once on the provided .txt files, inspect a record in Pinecone to confirm text and source metadata saved correctly



4. In Part 2, edit the Set Variable question text and Run once, confirm a relevant answer comes back





5. Test a question with no matching content (e.g. *"What's the weather in Melbourne?"*), confirm the model says it doesn't have enough information rather than hallucinating
6. Modify one parameter: **Limit** (top-k) in the Pinecone Search module, or **Chunk size** in Part 1
7. Re-run and compare results before/after

Checkpoint questions:

- Which chunks did the vector DB retrieve for your question, and were they actually relevant?
- What happened when you asked a question with no matching content?
- What changed after your modification?

Deliverable: Screenshot of a Pinecone record showing stored metadata, screenshot of a Part 2 run showing the final answer, plus a short note on the modification and its effect.